Task 1: Introduction to Artificial Intelligence

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1. **What is Artificial Intelligence?**

Artificial Intelligence (AI) is a branch of Computer Science that enables machines to perform tasks that normally require human intelligence — such as reasoning, learning, problem-solving, and decision-making. Instead of storing fixed answers, AI models are trained on large amounts of data to recognize patterns, and use this learning to generate responses or make predictions for new situations. Tools like ChatGPT and Claude are examples of AI — they can answer questions, solve reasoning problems, and even assist with aptitude-style questions by applying what they've learned during training.

2. **Types of AI (ANI, AGI, ASI)**

ANI (Artificial Narrow Intelligence) — "Weak AI": Designed to perform one specific task well, but cannot go beyond it. Examples: Siri, Alexa (voice commands only), ChatGPT (text-based tasks), spam filters. Most AI that exists today is ANI.

AGI (Artificial General Intelligence) — "Strong AI": Hypothetical AI that would have human-like general intelligence — able to learn, reason, and adapt across *any* task, not just one it was specifically trained for, similar to how a human can learn new skills. AGI does not fully exist yet; it's a future goal researchers are working toward.

ASI (Artificial Super Intelligence): AI that would surpass human intelligence in all areas — including creativity, reasoning, and problem-solving — such as predicting complex systems like climate patterns beyond human capability. Like AGI, ASI is still theoretical and doesn't exist yet.

3. **AI vs Machine Learning vs Deep Learning**

AI (the outermost, broadest concept): The overall goal — machines performing tasks that require human-like intelligence. This includes many approaches, not just machine learning (e.g., rule-based systems like your Task 5 chatbot are also technically AI, just not ML).

Machine Learning (a subset of AI): Instead of being explicitly programmed with fixed rules for every situation, ML systems **learn patterns from data** and improve their performance over time. For example, an ML model can learn to detect spam emails by studying thousands of examples, rather than someone manually writing rules for every possible spam pattern.

Deep Learning (a subset of Machine Learning): Uses **neural networks** — layered structures loosely inspired by how the human brain processes information — to learn from very large amounts of data. Deep Learning powers things like image recognition, voice assistants, and large language models like ChatGPT and Claude.

4. Real-life applications of AI

Virtual Assistants — Siri, Alexa, and Google Assistant use AI to understand voice commands, set reminders, and control smart devices.

Recommendation Systems — Netflix and YouTube use AI to analyze viewing history and suggest content tailored to each user.

Chatbots & AI Assistants — Tools like ChatGPT and Claude use AI to understand questions and generate helpful answers. I've also built a similar AI-powered chatbot myself — CampusNova — which uses RAG (Retrieval Augmented Generation) to answer student queries from college documents.

Face Recognition — Used in phone unlocking and security systems to identify individuals from facial features.

Self-Driving Cars — Companies like Tesla use AI to process real-time camera and sensor data to detect obstacles and make driving decisions.

5. Advantages and Limitations of AI

Advantages:

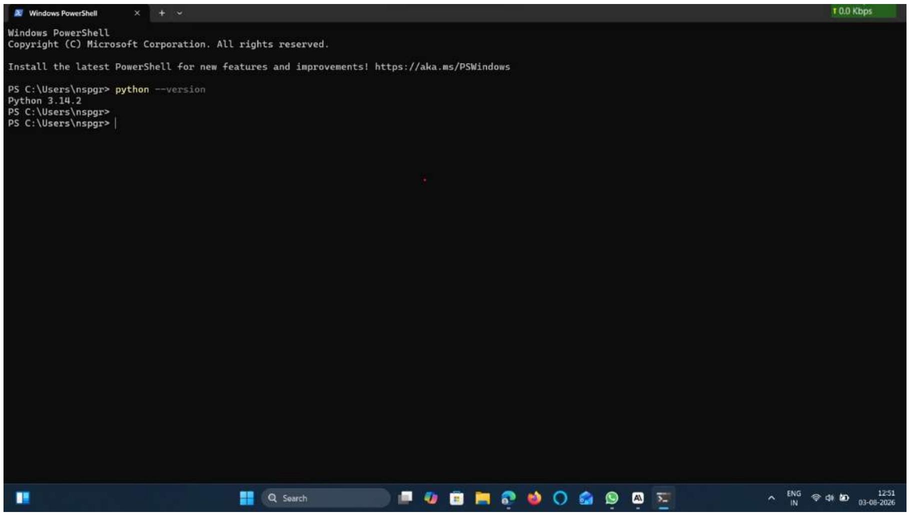
- Performs repetitive tasks faster and more accurately than humans
- Available 24/7 without fatigue
- Can process huge amounts of data quickly to find patterns humans might miss
- Reduces human error in tasks like calculations and data analysis

Limitations:

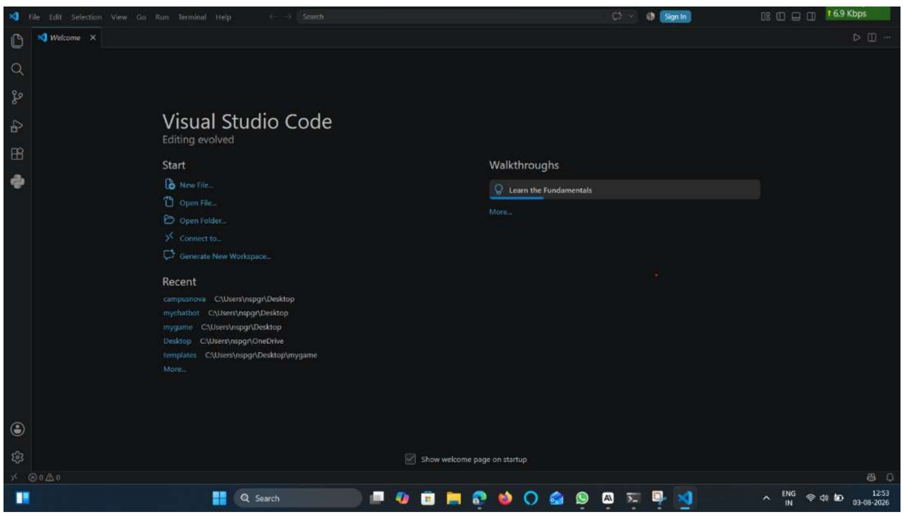
- Can produce incorrect information confidently (called "hallucination")
- Requires large amounts of quality data to work well
- Lacks true understanding or common sense — it recognizes patterns, not meaning
- Raises concerns around job displacement and data privacy

Task 2: Setting Up AI Development Environment

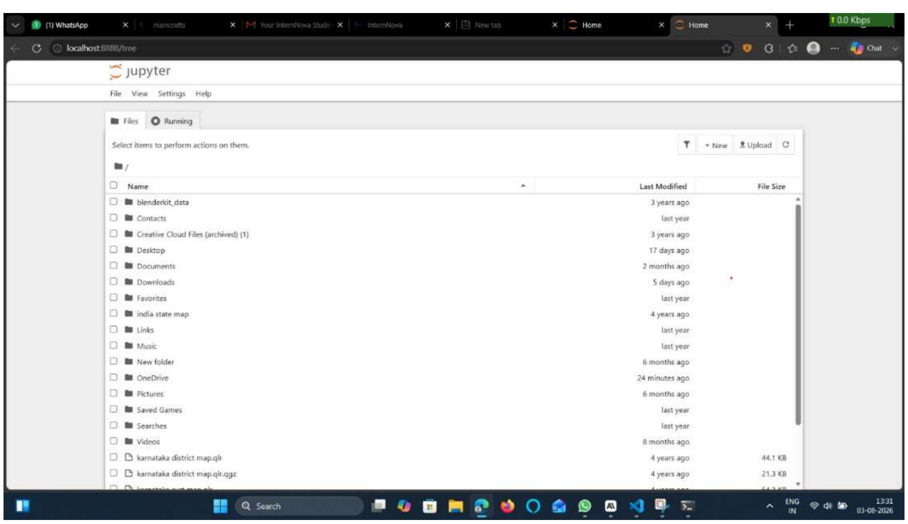
PYTHON



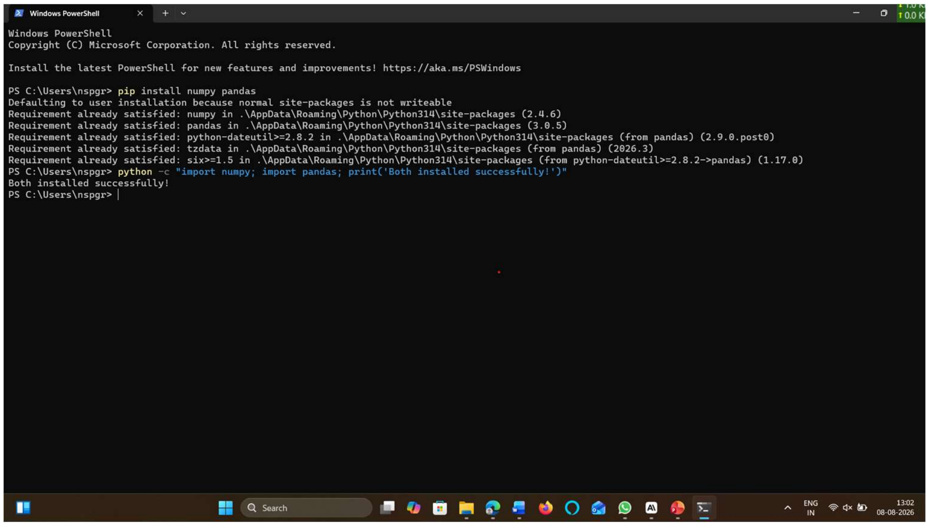
VSCODE



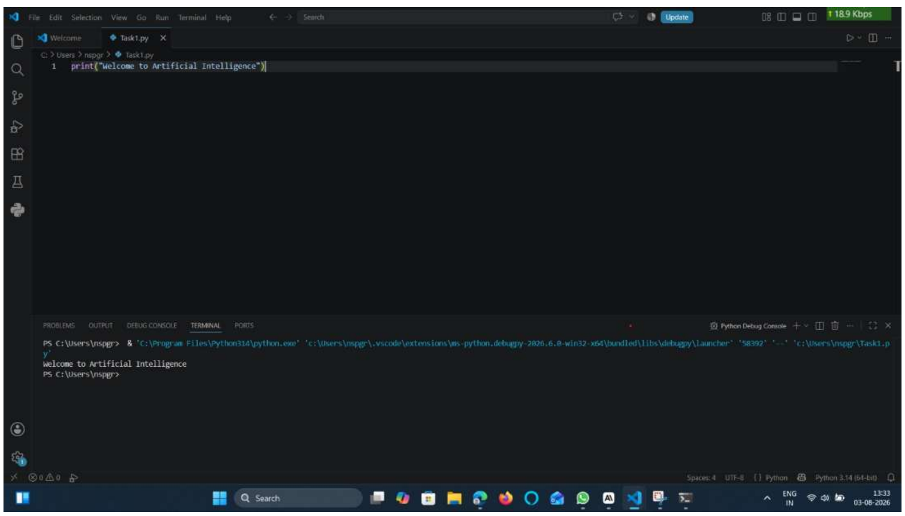
JYPUTER NOTEBOOK



Required libraries (NumPy, Pandas)



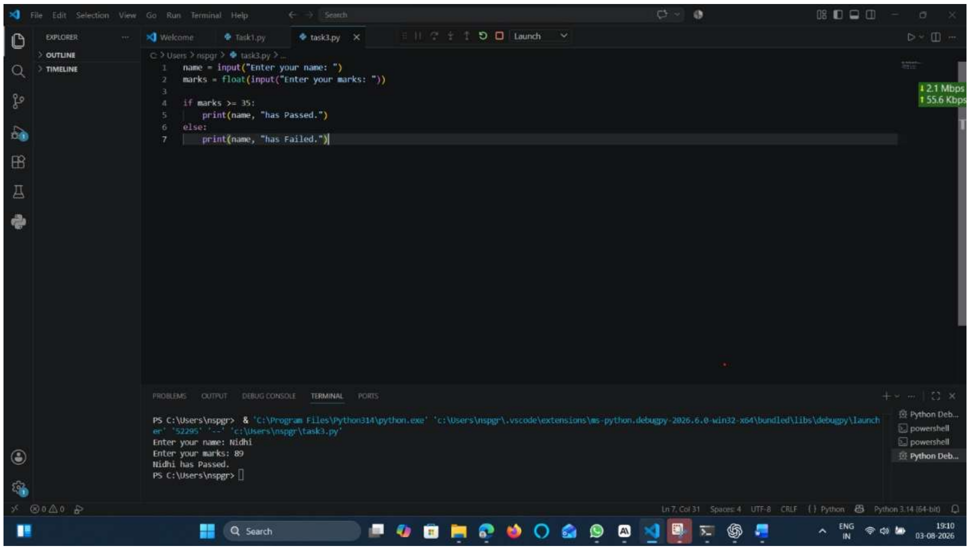
PROGRAM OUTPUT

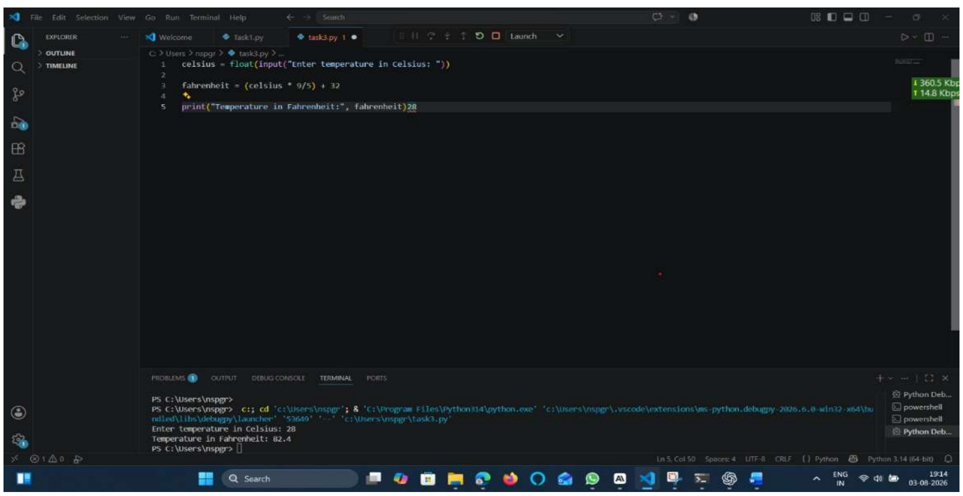


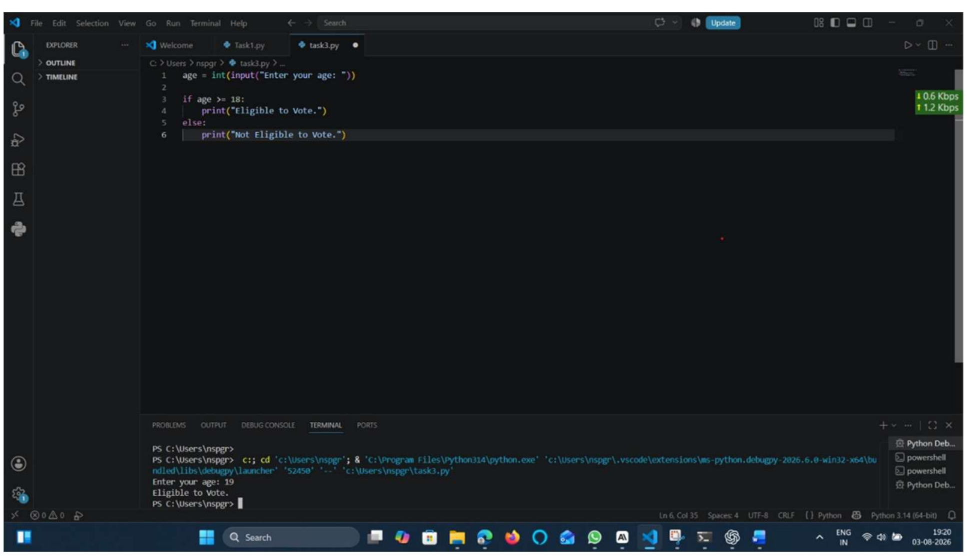
Task 3: Python for AI

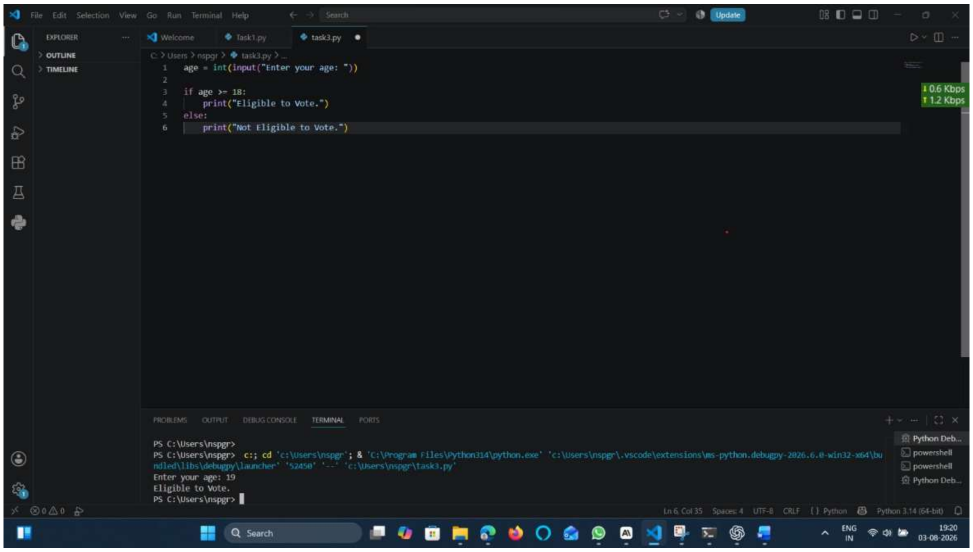
Write programs for: Variables, Data Types, User Input Operators.

Use an AI-related example such as: Predicting pass/fail based on marks Temperature conversion for a weather AI Age eligibility checker


name = input("Enter your name: ")
marks = float(input("Enter your marks: "))
if marks >= 35:
print(name, "has Passed.")
else:
print(name, "has Failed.")
Enter your name: Nidhi
Enter your marks: 89
Nidhi has Passed.


celsius = float(input("Enter temperature in Celsius: "))
fahrenheit = (celsius * 9/5) + 32
print("Temperature in Fahrenheit:", fahrenheit)
Enter temperature in Celsius: 28
Temperature in Fahrenheit: 82.4


age = int(input("Enter your age: "))
if age >= 18:
print("Eligible to Vote.")
else:
print("Not Eligible to Vote.")
Enter your age: 19
Eligible to Vote.



Task 4: AI Case Study

ChatGPT is an AI-powered chatbot developed by OpenAI that uses a large language model (LLM) to understand and generate human-like text. It is one of the most widely used AI applications today, helping people with writing, coding, research, and answering questions across almost any topic.

Input: The input to ChatGPT is a text-based question or instruction typed by the user — for example, asking it to explain a concept, write code, or summarize information. ChatGPT can also handle follow-up questions within the same conversation, using earlier messages as context.

Processing: Once the input is received, ChatGPT processes it using a neural network trained on massive amounts of text data. Rather than looking up pre-stored answers, it predicts the most likely and relevant sequence of words to generate as a response, based on patterns it learned during training. This is powered by deep learning and a transformer-based architecture, which allows it to understand context, grammar, and meaning across long pieces of text. In more advanced use cases, techniques like RAG (Retrieval Augmented Generation) can also be used, where the model retrieves relevant information from external documents before generating an answer — a technique I used myself while building CampusNova, an AI assistant for answering college-related student queries.

Output: The output is a natural-language response — text that directly answers the user's question, explains a concept, writes code, or completes whatever task was requested. The response is generated in real-time and displayed to the user almost instantly.

Benefits: ChatGPT saves significant time by answering questions instantly instead of requiring manual searching. It assists with learning, coding, writing, and problem-solving, making it useful for students, professionals, and researchers alike. It's available 24/7, making it more accessible than waiting for human assistance. Additionally, it helps break down complex topics into simple, understandable explanations, which is valuable for students trying to learn new concepts quickly.

However, it's worth noting a limitation: ChatGPT can sometimes generate incorrect information confidently (a phenomenon called "hallucination"), so its answers should be verified, especially for factual or technical accuracy.

Task 5: Mini AI Project